

27	B	When rotational speed of the a.c. generator is halved (i.e. angular speed ω is halved), the maximum induced e.m.f. $\varepsilon = NBA\omega$ will be halved. So, the peak induced current will be halved its initial value, i.e. $I_0 = 1 \text{ A}$. Mean power dissipated in the resistor $\langle P \rangle = I_{\text{r.m.s.}}^2 R$ $= \left(\frac{I_0}{\sqrt{2}} \right)^2 R$ $= \left(\frac{1}{\sqrt{2}} \right)^2 (20)$ $= 10 \text{ W}$
28	A	Kinetic energy of electron